1 The grammar

We use a formal grammar to define candidate circuits which we then fit on the data. The grammar we use does not impose any structural assumptions on the circuit. Each element can appear at any position, the only restriction being that a resistor `Rs` is in series with the rest of the circuit. The circuit elements present are standard equivalent circuit modeling (ECM) elements, which we present in Section 2.1 below.

$$\begin{aligned} \textit{Circuit} &\longrightarrow \texttt{Rs} \mid \texttt{Rs} + \textit{Network}, \\ \textit{Network} &\longrightarrow \textit{Element} \\ &\longrightarrow \textit{Network} + \textit{Network} \\ &\longrightarrow \textit{Network} \mid\mid \textit{Network}\ , \\ \textit{Element} &\longrightarrow \texttt{R} \mid \texttt{L} \mid \texttt{CPE} \mid \texttt{W}. \end{aligned} \tag{1}$$

2 Methodology

Several steps were taken in circuit generation, subsequent derivation of impedance equations and finally parameter fitting. Here we describe the core procedure.

2.1 Circuit generation

Circuit generation from formal grammars feels almost like solving a puzzle backwards. We consequently connect elements in series or parallel and in the end map each *Element* to either a resistor, inductor, Warburg element or a CPE. We replace two resistors (inductors) in series or in parallel with a single resistor (inductor), which helps to keep the total number of circuits down.

2.2 Impedance equation generation

Obtaining impedance equations from circuits generated by the grammar is straightforward. For series of elements we concatenate their respective impedances, and for elements in parallel we sum their admittances. Below we list the elements present in the grammar and their impedances (admittance is the inverse of impedance).

Constant phase element (CPE)

Constant phase element is a component that models an imperfect capacitor. It contains a dimensionless parameter $0 \leq \alpha \leq 1$ that can be interpreted as a measure of non-ideality. When $\alpha = 1$ a CPE is a perfect capacitor. CPE's impedance is

$$Z_{CPE} = \frac{1}{(j\omega)^{\alpha} Q}$$

where j is the imaginary unit, and Q is a parameter independent of the frequency. $\omega = 2\pi f$ is the angular frequency

Inductor (L) and Resistor (R)

Resistor's impedance is constant does not depend on the frequency, and inductor's impedance is a linear function of the frequency.

$$Z_R = R,$$

where R is the resistor's resistance, and

$$Z_L = j\omega L,$$

with L the inductance.

Warburg element (W)

Warburg element is used mainly in batteries EIS analysis. It is a special case of the CPE where $\alpha = 0.5$. We focused on the infinite Warburg element with impedance defined as

$$Z_W = \frac{\sigma}{\sqrt{jw}}$$

2.3 Parameter fitting

As described above, frequency domain impedance equations are directly generated from the circuits by simply summing up elements impedance formulas when they are connected in series, and summing up their reciprocal values when they are connected in parallel circuits.

We set a ceiling of the number of elements that may feature in the circuits. We run the optimization algorithm (more on that in a later section) up to a certain threshold, and for the circuits within a certain cutoff margin we re-run the optimization algorithm with stricter thresholds to obtain results as close to full convergence as possible.

We sort these 5 based on standard BIC to include some penalty for circuit complexity.

2.3.1 Initial value selection

We opted for a multi start stratified approach. Each circuit gets fitted 6 times in the initial evaluation. First fit is using deterministic fixed values p_{fixed}:

- 0.8 for α
- 0.05 for R_s
- 0.02 for other resistors R
- 1.0 for Q
- 0.01 for σ

Parameter	value
`x_scale`	`jac`
`ftol`	10-6
`xtol`	10-6
`gtol`	10-6
`f_scale`	0.001
`max_nfev`	100
`loss`	`soft_l1`
`method`	`trf`

Table 1: Parameters for the SciPy.optimize least_squares algorithm in the first optimization

- 10^{-6} for L

For the remaining 5 starts we selected the values as follows:

- α: Divide interval $[0.03, 0.98]$ in 5 equal parts and sample from each of the subintervals
- R_s: Divide interval $[0.05 \cdot 10^{-1}, 0.05 \cdot 10] = [0.005, 0.5]$ into 5 parts and sample from subintervals
- For the other parameters: Divide $[p_{fixed} \cdot 10^{-2}, p_{fixed} \cdot 10^{2}]$ in 5 subintervals and sample from them

This way we get a decent spread of initial values. We chose a narrower band for R_s as it is more directly constrained by the higher frequencies, and kept a broader band to allow for more uncertainty in initial values.

2.3.2 Numerical algorithm

In order to improve results of the algorithm we tried to improve robustness of SciPy's `least_squares` algorithm. For the initial optimization run we use:

TRF method means we use the *Trust region field* method of nonlinear least squares, and JAC scale is a specific way of calculating matrix column scales. Tolerance fields (f-,x-,g-) stop the iteration if function values, parameters or function's derivatives move less than the threshold. Soft L1 loss with corresponding f-scale parameter means that instead of minimizing squared residuals, the optimizer is instead minimizing the following objective function:

$$F(\boldsymbol{x}) = \frac{C^2}{2} \sum_{i=1}^{m} \rho\left(\frac{r_i(\boldsymbol{x})^2}{C^2}\right), \tag{2}$$

where $r_i(\boldsymbol{x})$ denotes the ith residual. For the `soft_l1` loss with `f_scale = 0.001`,

$$\rho(z) = 2\left(\sqrt{1+z} - 1\right), \qquad C = 10^{-3}. \tag{3}$$

Simply put, more residuals behave quadratically with a larger f-scale, while a smaller f-scale means most of the function is behaving like the L1 loss. For the first optimization round, so

Parameter	value
`x_scale`	`jac`
`ftol`	10-8
`xtol`	10-8
`gtol`	10-8
`f_scale`	0.001
`max_nfev`	10000
`loss`	`soft_l1`
`method`	`trf`

Table 2: Parameters for the SciPy.optimize least_squares algorithm in the second optimization round

for all circuits generated, the stopping point was the 100 function evaluations.

Substituting these expressions gives the optimization problem

$$\boxed{\min_{\boldsymbol{x}} F(\boldsymbol{x}) = \min_{\boldsymbol{x}} 10^{-6} \sum_{i=1}^{m} \left[\sqrt{1 + 10^6 r_i(\boldsymbol{x})^2} - 1 \right].} \tag{4}$$

In the second optimization round, that is, for 5 circuits with smallest MSE values, we again used the same algorithm, but with stricter optimization values presented in Table 2. Main difference is the increased number of steps and stricter function evaluation. In this case, for each of the circuits one of the tolerances was reached before we hit the step limit.

2.3.3 Normalising residuals

We obtain one scale value per frequency in either of the two types of data used. For SOFC data, since each dataset is comprised of several measurements across 233 frequencies, we take a median of impedances. This leaves us with a 233 point dataset. To calculate the scales, we take the median absolute deviation (MAD) of impedances of each frequency.

For the battery data, we first need to derive the impedances from the raw data available. Since there are many datapoints available we randomly sample 300 points for the data

We noticed that in some cases the scales can be quite small, which makes dividing residual values (which can also be small) with them unstable. To this end, we replace the smallest 10 % of the values of the scale with a fixed value. It should be noted that we calculate separate scale values for imaginary and real parts of the impedances.

3 Results

We apply methods described above to data from both solid oxide fuel cells, as well as batteries. We try to distinguish which of the numerical adaptations of the algorithm (additional settings for the SciPy's least-squares solver, normalising residuals) have the largest impact on the fit.

The plots presented in the graphs are the Nyquist graphs of the 5 circuits which we fitted

until convergence, ordered by their BIC values.

3.1 SOFC data

We took `eis_20200227_070006.npz` dataset from repository https://portal.ijs.si/nextcloud/s/xTa2cmtfxXn2jSz. As mentioned in Section 2.3.3, we have several repeated measurements of impedance at each frequency. We take the median of them to get one impedance value per frequency.

In Figure 1 we see results of fitting circuits with up to 4 elements. They do not capture any of the details of the data.

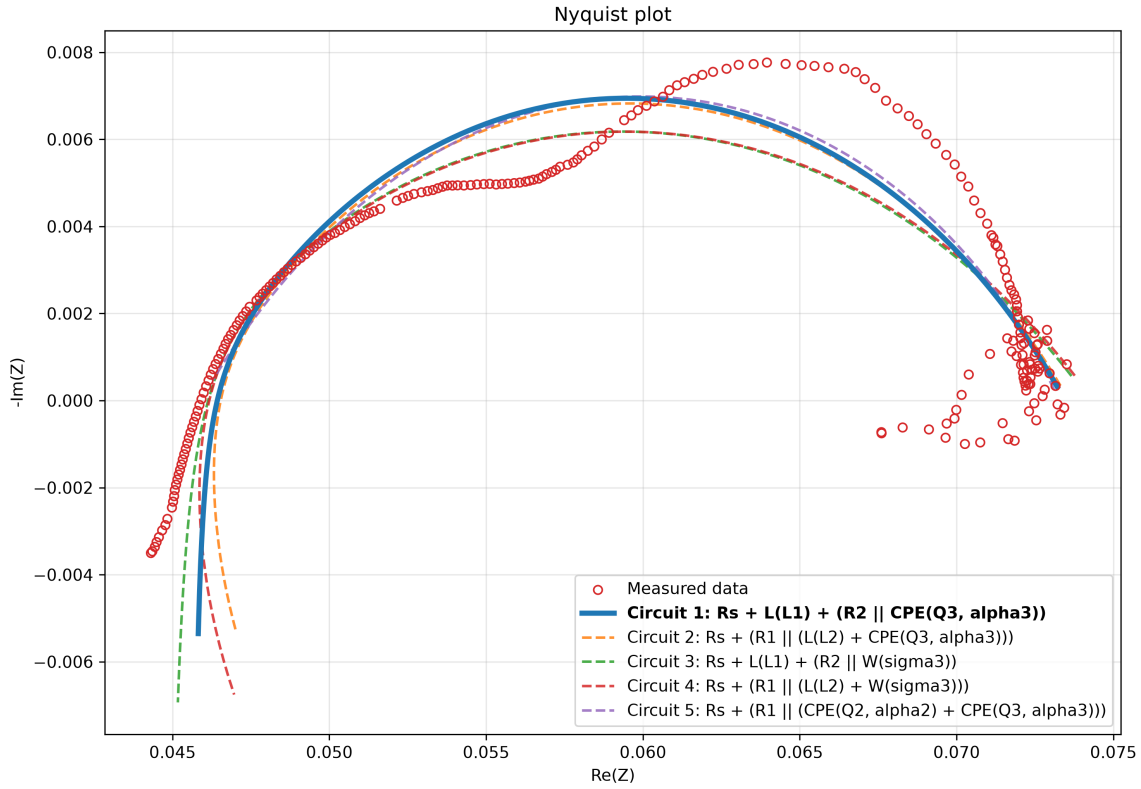


Figure 1: Nyquist graphs of 5 best fitting circuits according to our algorithm with up to 4 elements

Results with an additional circuit element are only marginally better (Figure 2).

Before expanding the search space with even more circuits, we decided to try to make the algorithm more robust. We first implemented normalising residuals at each step of the algorithm and present results in Figure 3. There is noticable improvement in how closely the graphs fit the data. When inspecting the results in Table 5 more closely, we see that several parameters are close to their limits (especially α).

To improve on unstable solutions we robustify the algorithm with adaptations described in Section 2.3.2. This way we get some more diversity in results in Graph 4.

Comparing the circuits obtained with the standard and robustified numerical procedures we

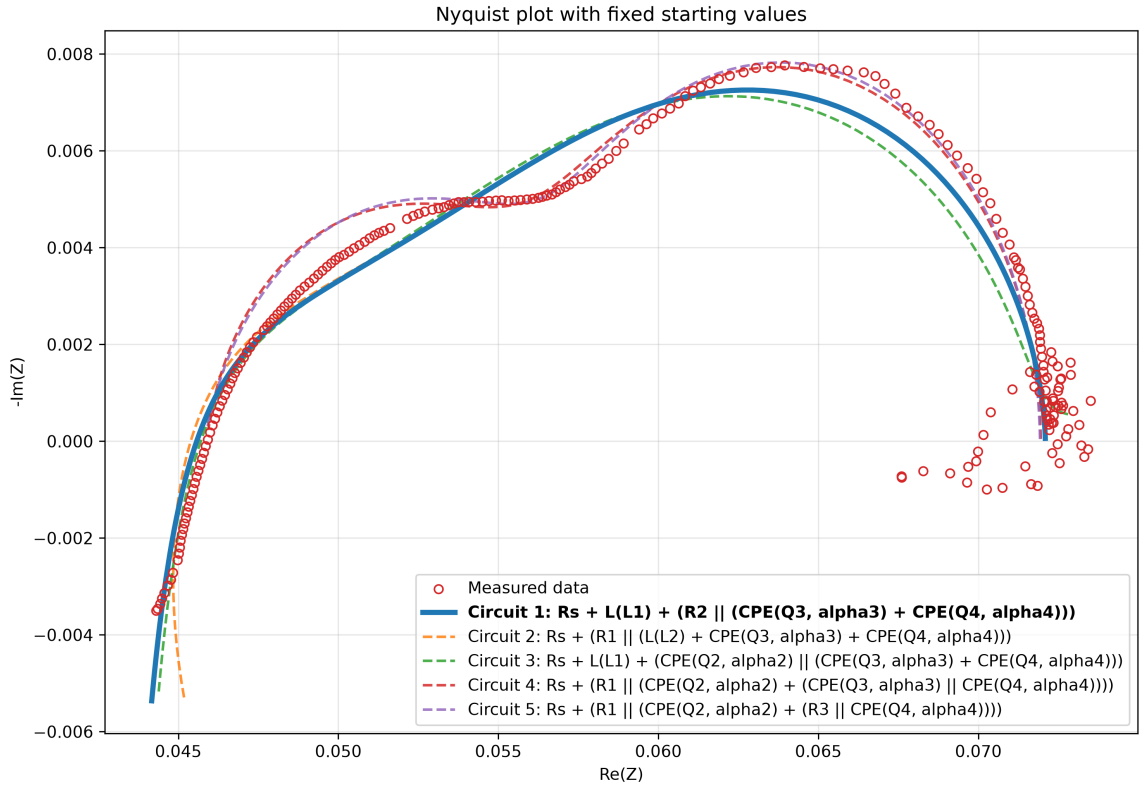


Figure 2: Nyquist graphs of 5 best fitting circuits according to our algorithm with up to 5 elements

get can see that the former contains one parameter less (but same number of elements), and none of the parameters are on the border of their permitted range. MSE difference between them was minimal. The circuits in question were:

$$\text{Robustified approach} = R_s \oplus [\text{CPE}(Q_1, \alpha_1) \parallel (R_2 \oplus [R_3 \parallel \text{CPE}(Q_4, \alpha_4)])]\,,$$

$$\text{Standard approach} = R_s \oplus [R_1 \parallel (\text{CPE}(Q_2, \alpha_2) \oplus [\text{CPE}(Q_3, \alpha_3) \parallel \text{CPE}(Q_4, \alpha_4)])]\,.$$

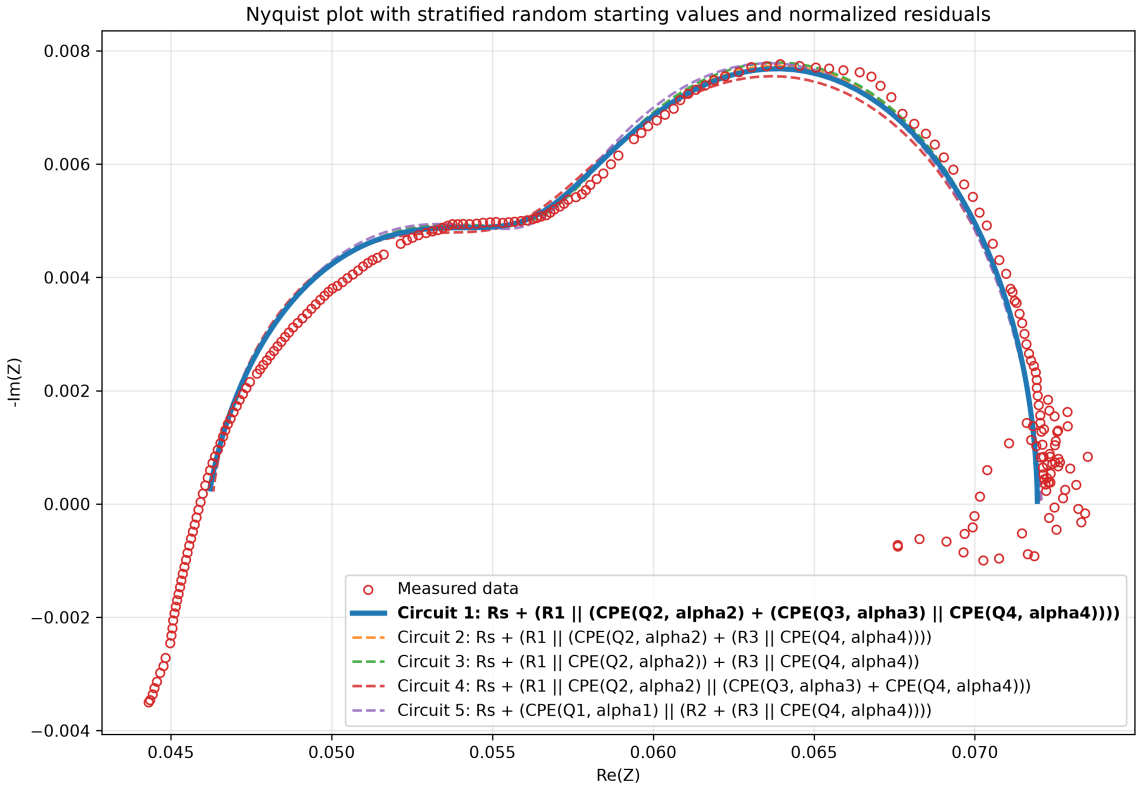


Figure 3: Nyquist graphs of 5 best fitting circuits according to our algorithm with up to 5 elements and normalised residuals.

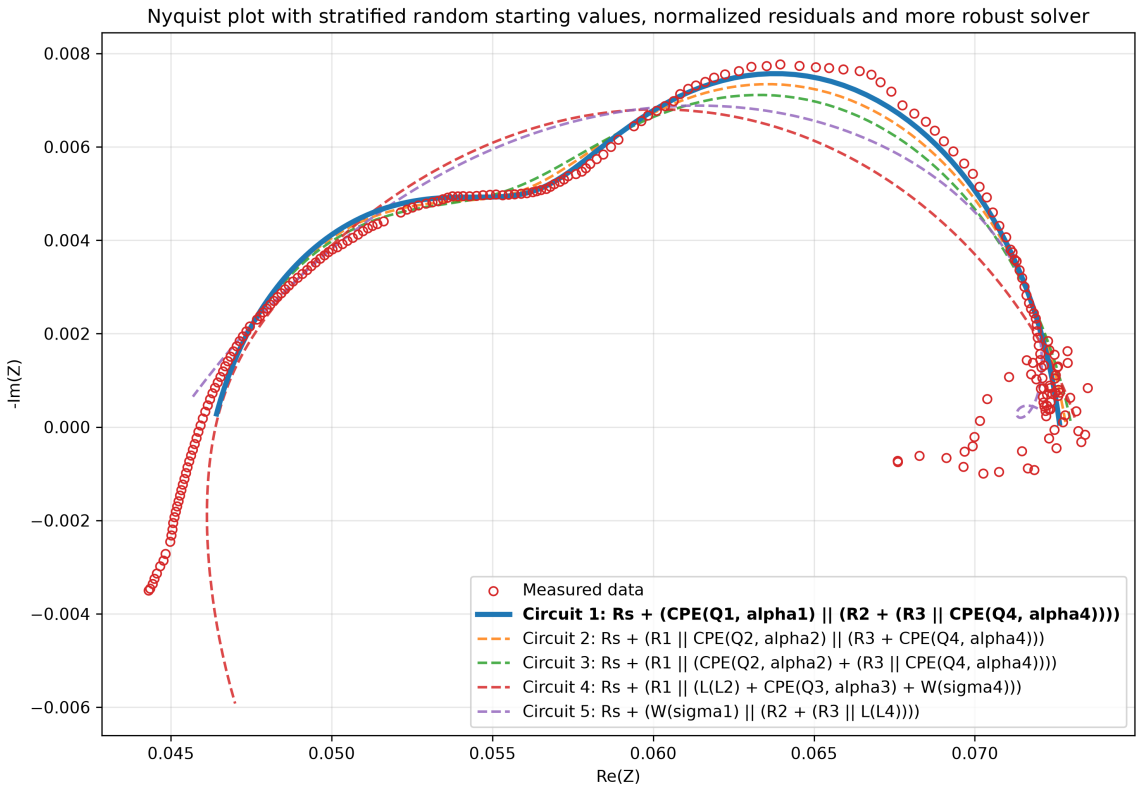
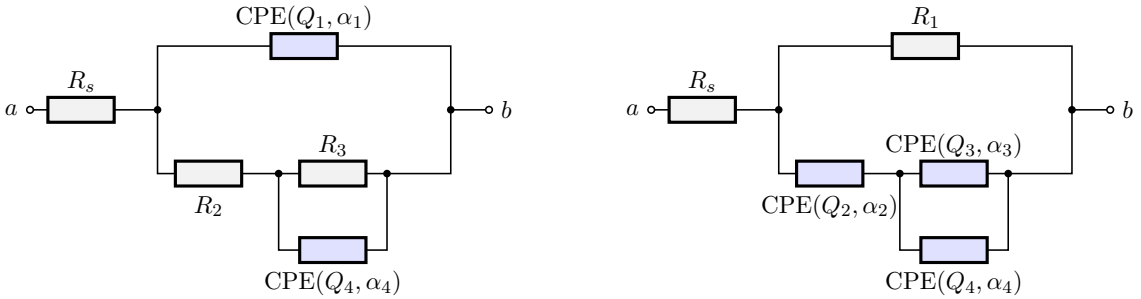


Figure 4: Nyquist graphs of 5 best fitting circuits according to our algorithm with up to 5 elements and robustified algorithm



(a) Result of the robustified approach

(b) Result of the standard approach

Figure 5: Resulting circuits from two approaches with parameters described in 3: (a) robustified algorithm; (b) standard algorithm.

Table 3: Parameter comparison of the two circuits

Parameter	**Robustified algorithm**	**Standard approach**
Element count	5	5
Parameter count	7	8
MSE	1.89647×10^{-6}	1.64223×10^{-6}
BIC	-3031.73789	-3059.82487
Circuit parameters		
R_s (Ω)	0.0463228	0.0461368
R_1 (Ω)	—	0.0258082
R_2 (Ω)	0.0129929	—
R_3 (Ω)	0.0133333	—
Q_1 ($\mathrm{S\,s}^{\alpha_1}$)	0.542887	—
α_1	0.795783	—
Q_2 ($\mathrm{S\,s}^{\alpha_2}$)	—	1.05020
α_2	—	0.990000
Q_3 ($\mathrm{S\,s}^{\alpha_3}$)	—	50.8970
α_3	—	0.0298022
Q_4 ($\mathrm{S\,s}^{\alpha_4}$)	3.33826	0.469656
α_4	0.941526	0.819800

3.2 Battery data

We apply the methods to data from batteries as well. We used data from https://portal.ijs.si/nextcloud/s/2TRZ

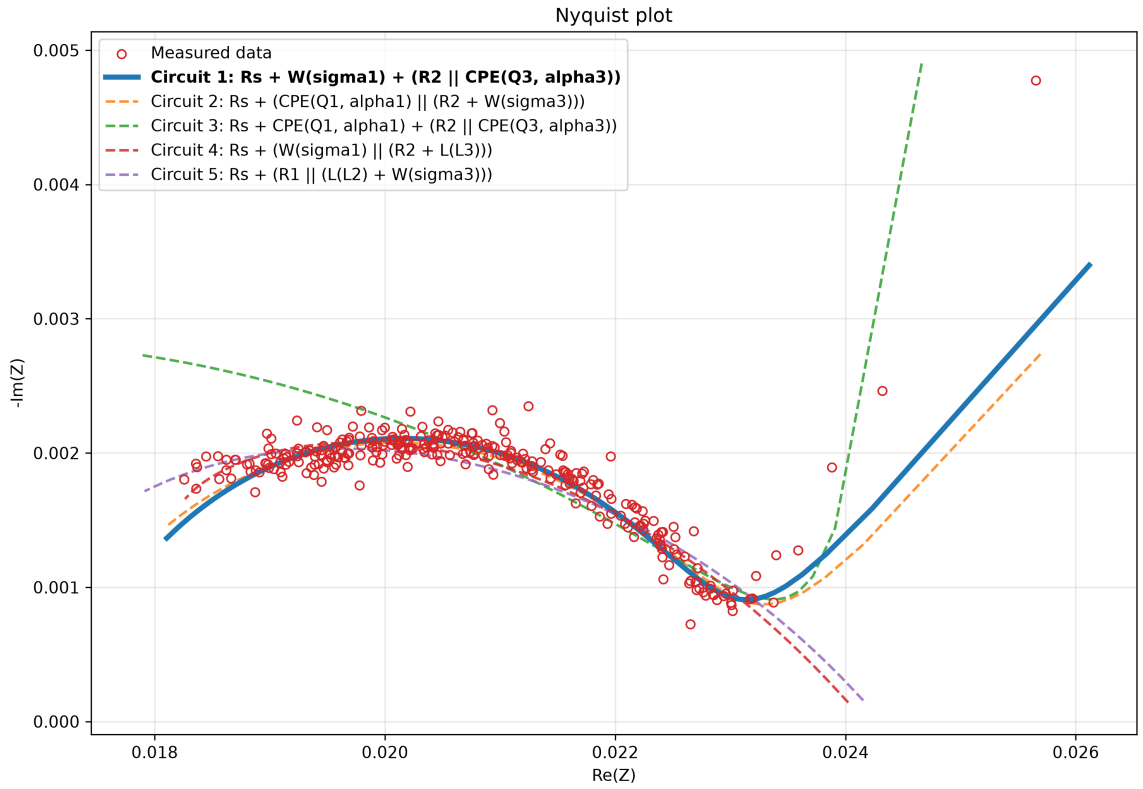


Figure 6: Nyquist graphs of 5 best fitting circuits according to our algorithm with up to 4 elements

We compare the 5 element circuits obtained from robustified and standard numerical algorithm.

$$\text{Robustified approach} = R_s \oplus \left[(\text{CPE}(Q_1, \alpha_1) \oplus \text{CPE}(Q_2, \alpha_2)) \parallel (R_3 \oplus \text{CPE}(Q_4, \alpha_4))\right],$$

$$\text{Standard approach} = R_s \oplus \text{CPE}(Q_1, \alpha_1) \oplus \left[R_2 \parallel (\text{CPE}(Q_3, \alpha_3) \oplus \text{CPE}(Q_4, \alpha_4))\right].$$

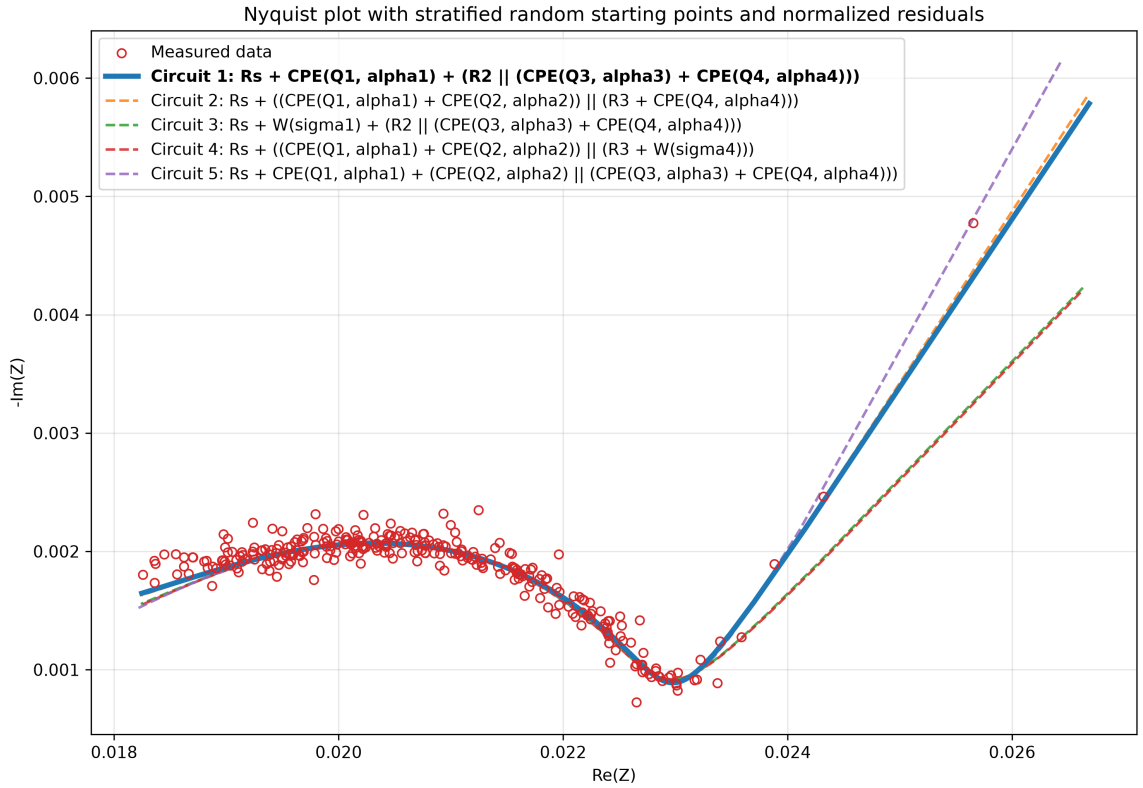


Figure 7: Nyquist graphs of 5 best fitting circuits according to our algorithm with up to 5 elements

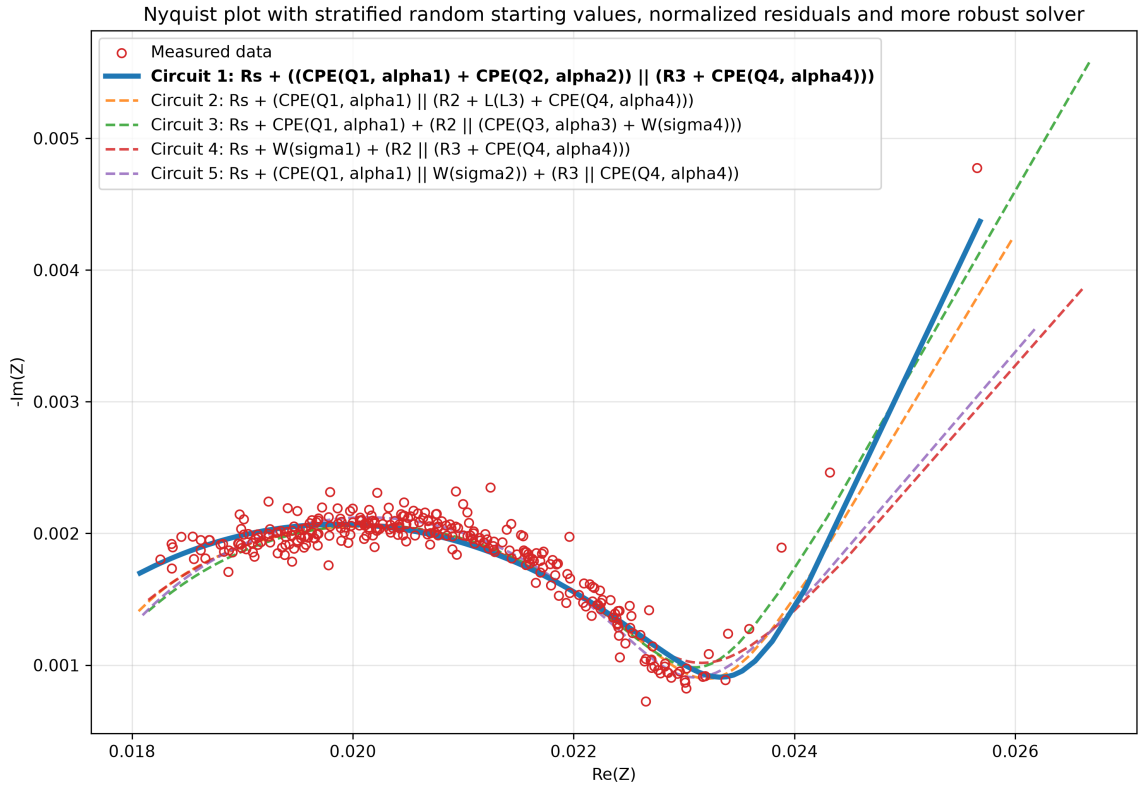
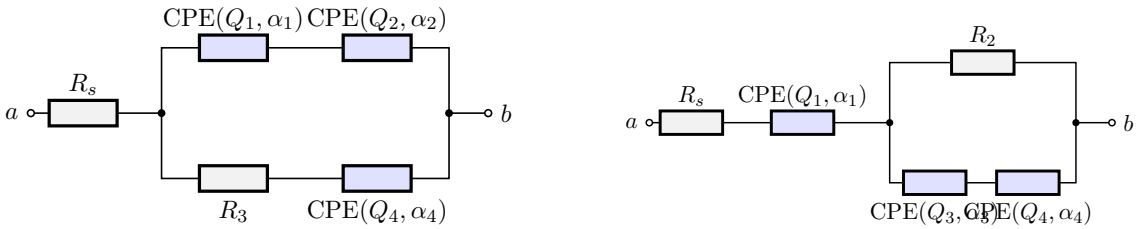


Figure 8: Nyquist graphs of 5 best fitting circuits according to our algorithm with up to 5 elements and robustified algorithm



(a) Extra-numerics fit (b) Standard fit

Figure 9: Side by side cirucit comparison: (a) Robustified approach (b) Standard approach

Table 4: 5 element circuit parameter comparison

Parameter	**Robustified approach**	**Standard approach**
Element count	5	5
Parameter count	8	8
MSE	2.72147×10^{-8}	2.33564×10^{-8}
BIC	-5180.22	-5226.09
Circuit parameters		
R_s (Ω)	0.0160873	8.08927×10^{-5}
R_2 (Ω)	—	0.0225741
R_3 (Ω)	0.00747417	—
Q_1 ($\mathrm{S\,s}^{\alpha_1}$)	122.545	505.257
α_1	0.664787	0.612086
Q_2 ($\mathrm{S\,s}^{\alpha_2}$)	5.36106	—
α_2	0.640431	—
Q_3 ($\mathrm{S\,s}^{\alpha_3}$)	—	4.29185
α_3	—	0.166491
Q_4 ($\mathrm{S\,s}^{\alpha_4}$)	859.752	0.122094
α_4	0.693547	0.989999

4 EIS Battery data

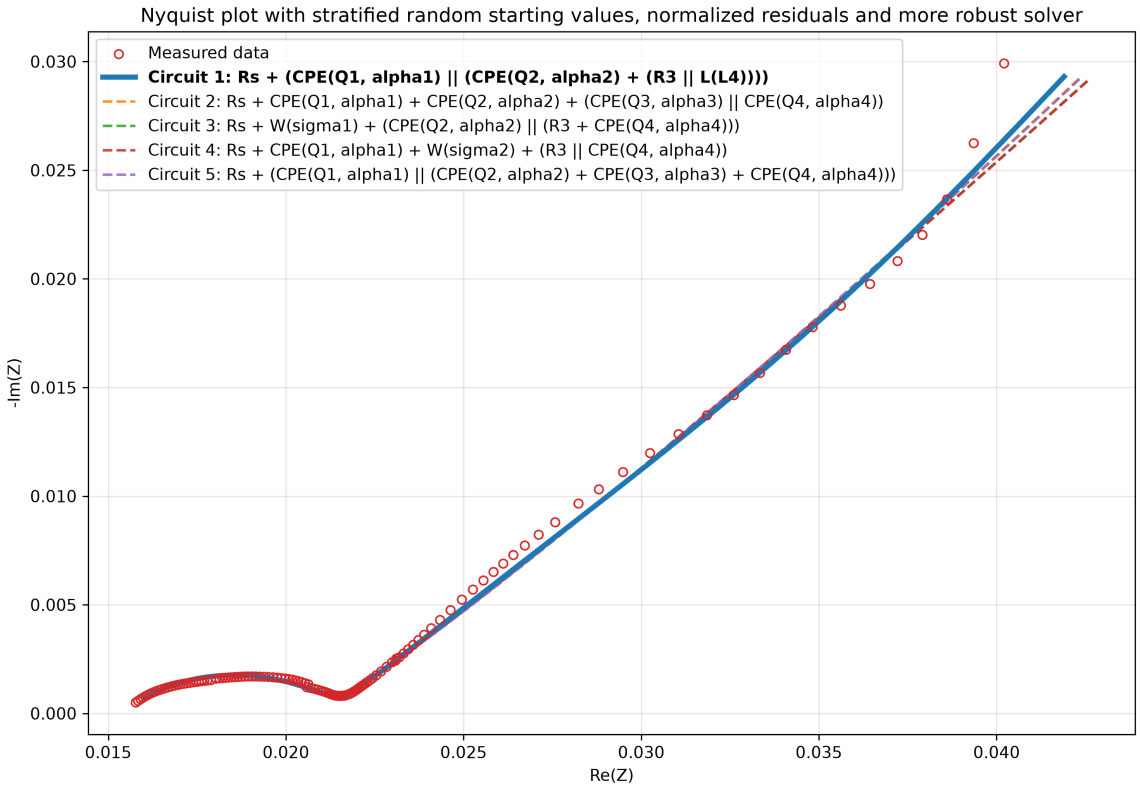


Figure 10: Nyquist graphs of 5 best fitting circuits according to our algorithm with up to 5 elements and robustified algorithm

Metric	**Value**
Element count	5
Parameter count	7
MSE	1.49010×10^{-7}
BIC	-3985.31221
Circuit parameter	
R_s (Ω)	0.0155126
R_3 (Ω)	0.00593952
L_4 (H)	0.734682
Q_1 ($\mathrm{S\,s}^{\alpha_1}$)	2.91787
α_1	0.673854
Q_2 ($\mathrm{S\,s}^{\alpha_2}$)	433.414
α_2	0.602733

Table 5: Fitted parameters of the best-performing circuit.

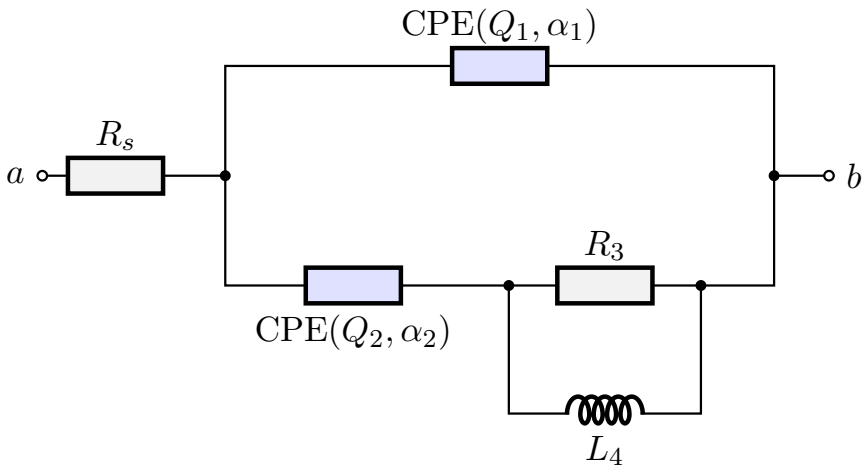


Figure 11: Best-performing equivalent circuit with parameters listed in Table 5.

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